CS23334 Fundamentals of Data Science Lab

Experiment 20

Exp No: 20 Experiment to understand K-means clustering algorithm for a given data set.

Description: Understand the K-means clustering algorithm for the dataset given.

Aim: To build, train, evaluate, and visualize an Unsupervised K-Means Clustering model on Mall_Customers.csv to segment customers based on Annual Income, Spending Score, and Age attributes.

Methodology:

1 **Data Acquisition & Inspection:** Load Mall_Customers.csv containing 200 customer entries with 5 features (CustomerID, Gender, Age, Annual Income (k$), Spending Score (1-100)). Audit dataset schema via df.info().

2 **Exploratory Pairwise Plotting:** Generate a 4x4 feature scatter grid using sns.pairplot(df) to visually identify natural clustering tendencies.

3 **2D Feature Extraction & K-Means Fitting:** Extract Annual Income (k$) and Spending Score (1-100) into features matrix (df.iloc[:, [3,4]].values). Fit KMeans(n_clusters=5) and assign cluster labels to Final DataFrame.

4 **Cluster Segmentation Plotting:** Map cluster labels using sns.FacetGrid(Final, hue="label", height=8) to visualize 5 customer market segments.

5 **3D Feature Extraction & Elbow Method:** Extract Age, Annual Income (k$), and Spending Score (1-100) into features_el. Compute Within-Cluster Sum of Squares (WCSS/inertia_) for cluster counts $k = 1$ to 9.

6 **Elbow Curve Visualization:** Plot WCSS versus k using plt.plot(range(1,10), wcss) to determine the optimal number of clusters ($k=5$).

Code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

df=pd.read_csv('Mall_Customers.csv')

df.info()

df.head()

sns.pairplot(df)

features=df.iloc[:,[3,4]].values

from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
```

```
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()

sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
   .map(plt.scatter,"Annual Income (k$)","Spending Score (1-100)") \
   .add_legend();
plt.show()

features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

OUTPUT: 1. Dataset Overview & Pairwise Feature Exploration

```
In [1]: import numpy as np
        import pandas as pd
        import matplotlib.pyplot as plt
        import seaborn as sns
        %matplotlib inline

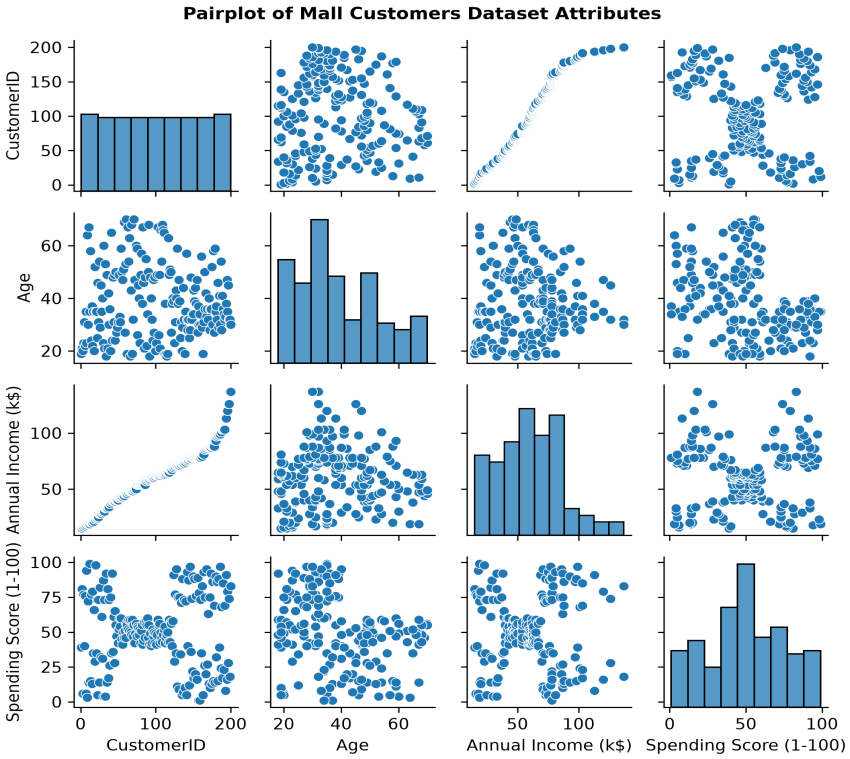
In [2]: df=pd.read_csv('Mall_Customers.csv')

In [3]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
In [4]: df.head()
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
In [5]: sns.pairplot(df)
Out[5]: <seaborn.axisgrid.PairGrid at 0x170e8e47850>
```



OUTPUT: 2. K-Means Fitting & Cluster Segmentation

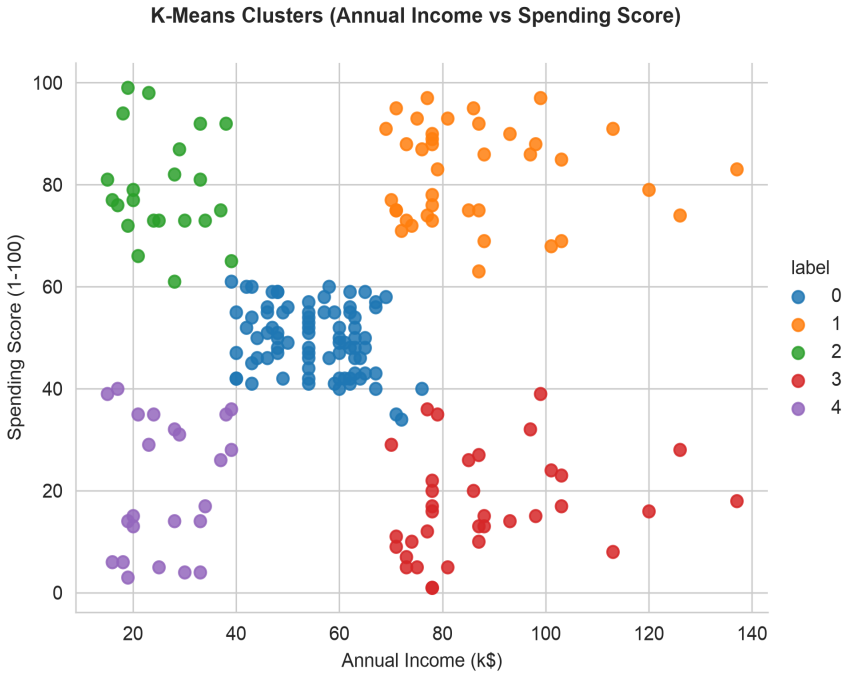
```
In [6]: features=df.iloc[:,[3,4]].values

In [7]: from sklearn.cluster import KMeans
        model=KMeans(n_clusters=5)
        model.fit(features)
Out[7]: KMeans(n_clusters=5)
```

```
In [8]: Final=df.iloc[:,[3,4]]
        Final['label']=model.predict(features)
        Final.head()
```

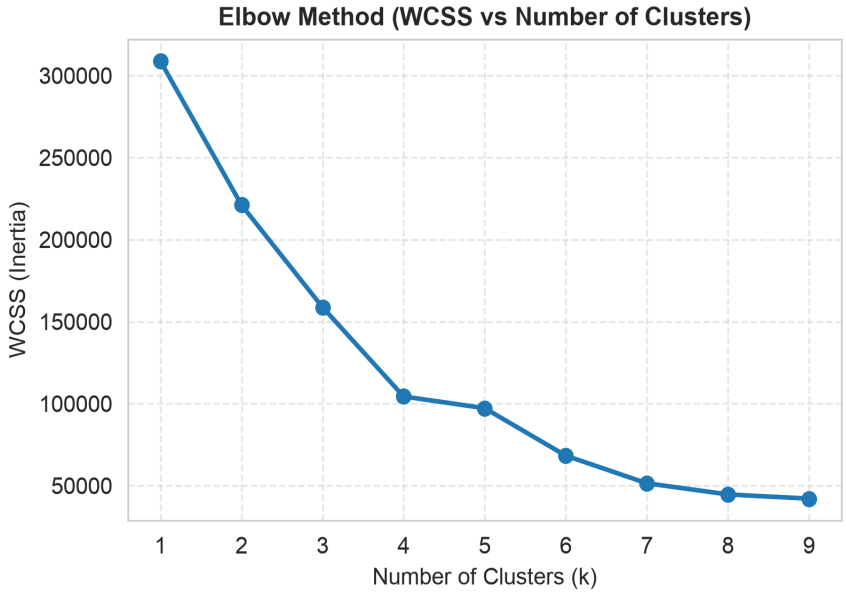
	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

```
In [9]: sns.set_style("whitegrid")
        sns.FacetGrid(Final,hue="label",height=8) \
           .map(plt.scatter,"Annual Income (k$)","Spending Score (1-100)") \
           .add_legend();
        plt.show()
```



OUTPUT: 3. Elbow Method Curve (WCSS Analysis)

```
In [10]: features_el=df.iloc[:,[2,3,4]].values
         from sklearn.cluster import KMeans
         wcss=[]
         for i in range(1,10):
             model=KMeans(n_clusters=i)
             model.fit(features_el)
             wcss.append(model.inertia_)
         plt.plot(range(1,10),wcss)
Out[10]: [<matplotlib.lines.Line2D at 0x170e99f3550>]
```



Result:

K-Means clustering on the Mall Customers dataset successfully identified 5 distinct customer segments based on Annual Income and Spending Score. The Elbow Method confirmed k=5 as the optimal cluster count. Scatter plots clearly visualized the clusters, enabling targeted marketing strategy recommendations.